

HAT-P-54b

R-1716

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_180314 · created 2026-08-02T16:05:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458191.63134 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.056
Transit depth (Rp/Rs)^2	3.26 +/- 2.03 [%]
Orbital Inclination (inc)	89.37 +/- 2.26
Airmass coefficient 1 (a1)	0.077 +/- 0.056
Airmass coefficient 2 (a2)	2.98 +/- 1.75
Scatter in the residuals of the lightcurve fit is	101.8 %
Best Comparison Star	#3 - [550, 75]
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.0518 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.181 $\pm$ 0.056 vs 0.1572 (deviation 0.42 σ)
Duration fit vs published	0.0518 d vs 0.075 d (deviation 7.03 σ)
Mid-time O-C	-0.6 min
Depth SNR	3.2
Residual scatter	101.8 %

Light curve

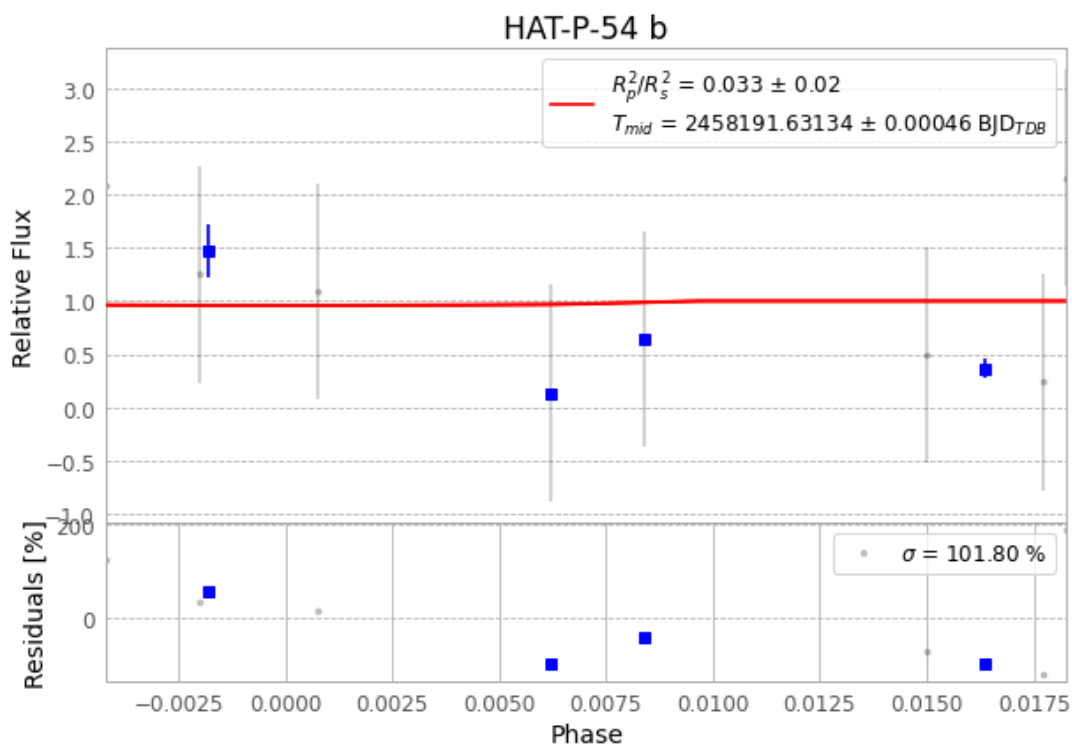


Figure 1 — FinalLightCurve_HAT-P-54 b_2018-03-13.png (SHA-256 a04c8b7ca6430b60...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 68], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aae3e0b831b16ee6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

48 FITS frames (32 MB), HATP-54180314023614.FITS → HATP-54180314050012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-180314.log (89fb2a309127...), FinalLightCurve_HAT-P-54 b_2018-03-13.png (a04c8b7ca643...), FinalLightCurve_HAT-P-54 b_2018-03-13.csv (e51d121a9356...)

Compute resources

Wall time 240.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:05Z.